

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++, Go, Java, Node.js, Kotlin, SQL

Tools/Frameworks: Docker, Kubernetes, TensorFlow, PyTorch, FastAPI, Flask, TensorRT, AWS SageMaker, OpenAI API, Airflow, PySpark, Hive, NiFi

Competence: Machine Learning, MLOps, Generative AI, LLMs, AI Model Deployment, Data Engineering, Cloud Architecture, API Development, Edge AI, Data Pipelines

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Developed and optimized deep learning models for metrological applications, enhancing model performance and training efficiency.
- Implemented novel architectures and physics-informed hybrid loss functions, improving model accuracy and stability.
- Collaborated with PhD researchers to co-author a publication presented at the FLOMEKO conference.
- Created interactive dashboards for real-time data visualization and model performance monitoring.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Increased vitiligo segmentation accuracy to 94% and reduced processing time by 5x using advanced AI algorithms.
- Developed a real-time tennis foul detection system with TensorRT, reducing compute costs and inference time by 6x.
- Designed and managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift and maintaining Kubernetes infrastructure.
- Led educational sessions on machine learning and big data for university students, fostering knowledge sharing and innovation.
- Built POCs using YOLO, few-shot learning, and CNNs for client demonstrations, showcasing AI capabilities.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed a full-stack web application for AI-based services using Flask, HTML, and CSS, enabling seamless user interaction.

- Built an AI chatbot leveraging OpenAI API and LLM tools, enhancing user engagement and automation capabilities.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Designed and implemented web scrapers and ETL pipelines for structured and unstructured data extraction and processing.
- Automated data ingestion and cleaning workflows to streamline client analytics and reporting processes.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability ensemble integrating DPFunc and DeepGOPlus with AlphaFold for feature extraction.
- Enhanced model accuracy by 4% through custom feature engineering for MF, CC, and BP namespaces.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a real-time Bitcoin price forecasting system using LSTM models, achieving high prediction accuracy.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Engineered the world's first AI-driven self-playing Chitralli Sitar, combining signal processing and music generation LLMs.
- Project featured on Coke Studio in a music video by Hassan Raheem, showcasing innovative AI applications in music.

Hugging Face

Open-Source

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in machine learning, generative AI, and scalable AI platform development.

National University of Computer and Emerging Sciences

Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor for academic excellence.
- Focused on data engineering, machine learning, and cloud computing.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python